

THE ENERGY AND RESOURCES INSTITUTE, DELHI BATTLING FOR EARTH

Electricity lines form a roof for the faraway Dakshin Dimoria village in the Kamrup district of Assam. Yet these lines of development don't reach 70 per cent of the households and for the rest, there is an erratic power supply. Villagers resort to using kerosene lamps and candles once dusk sets in. "One out of every four people in the world without access to electricity is an Indian," says R.K. Pachauri, director-general, The Energy and Resources Institute (TERI) and chairman, Intergovernmental Panel on Climate Change. Traditional fuels—firewood, animal dung and kerosene—still power rural India but also lead to adverse effects on health and carbon dioxide emissions. "We plan to bring five lakh solar lanterns to 5,000 villages by

the end of March 2009 under the Lighting a Billion Lives project," says Leena Srivastava, TERI executive director. It will eventually reduce 250 million tonne of carbon emissions, when measured in terms of kerosene utilisation avoided. TERI began in 1974, supporting research and enhancing the use of biogas technology but by 1982, with Pachauri taking over, refocused to building its own research facilities. It became the first institution in a developing country to give impetus to energy efficiency and renewables. "The motive behind TERI's activities is to move global society in general and India in particular towards a path of sustainable development," says Pachauri. TERI has grown fast—from a budget of Rs 3 crore, it is now a 750-

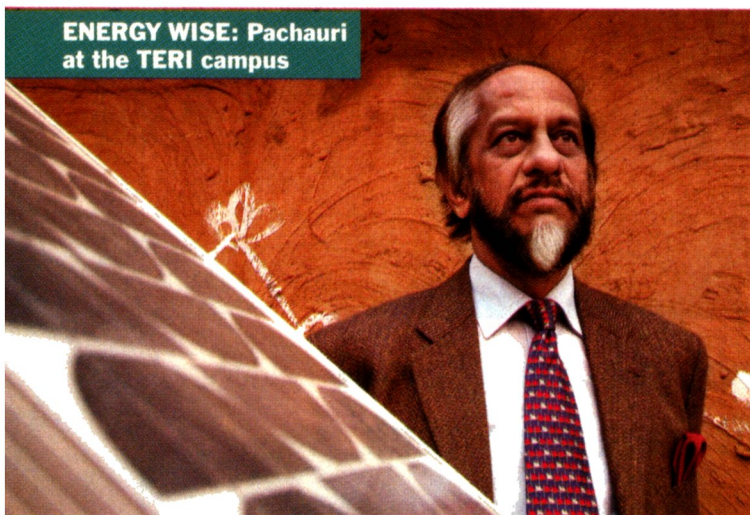
"TERI hopes to move global society and India towards sustainable policies."

R.K. PACHAURI, DIRECTOR-GENERAL, TERI

employee organisation with an approximate budget of Rs 750 crore, and has a presence in North America, Europe, Japan, Malaysia and the Gulf. It has pioneered the TERI-Griha—a green-rating system for integrated habitat assessment, which has been adopted by the Indian government, the Kyoto Protocol and the UN Framework Convention on Climate Change. With the TERI University functioning successfully, it now plans to open a management school for education on sustainability issues. With increasing worries over climate change, TERI's efforts to produce eco-conscious managers could not have come at a better time.

by Purvi Malhotra

ENERGY WISE: Pachauri at the TERI campus



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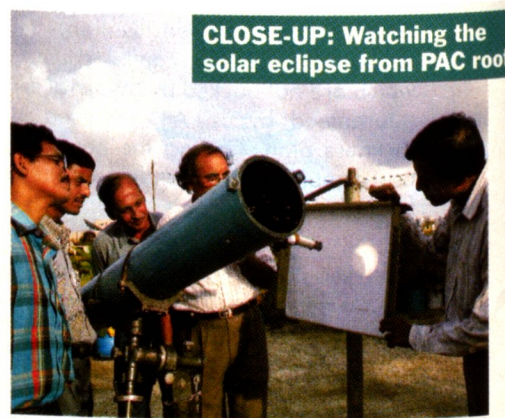
POSITIONAL ASTRONOMY CENTRE, KOLKATA SPEED DATES

India's diversity was not, and is still not, confined to language and culture alone. Till the 1950s, India had 33 types of calendars. This forced scientist Meghnad Saha to take up the matter with then prime minister Jawaharlal Nehru. "Saha wanted a unified calendar on the basis of accurate modern astronomical data for the sake of national integrity," says Amalendu Bandyopadhyay, the first director of the Positional Astronomy

Centre (PAC). The Calendar Reform Committee, with Saha as its chairman who travelled across the country collecting calendars, recommended the use of a solar calendar system for civil use and a luni-solar one for religious purposes. The PAC calculates the position of planets, lunar positions, tides and eclipses, which are used by soldiers, sailors as well as astronomers.

by Sumitro Bandyopadhyay

CLOSE-UP: Watching the solar eclipse from PAC roof



SUBIR HALDER

7 countries, including India, calculate the position of the planets for civil, nautical and astronomical purposes.